Sigmoid / Logistic Regression

From numerical predictions to classes and probabilities

Hand-calculation worksheet

Name _____ Date _____

01 Binary Classification

* Synthetic practice data Sensor signal x predicts status y Normal 0, abnormal 1

1 Write the possible target values and explain the positive class

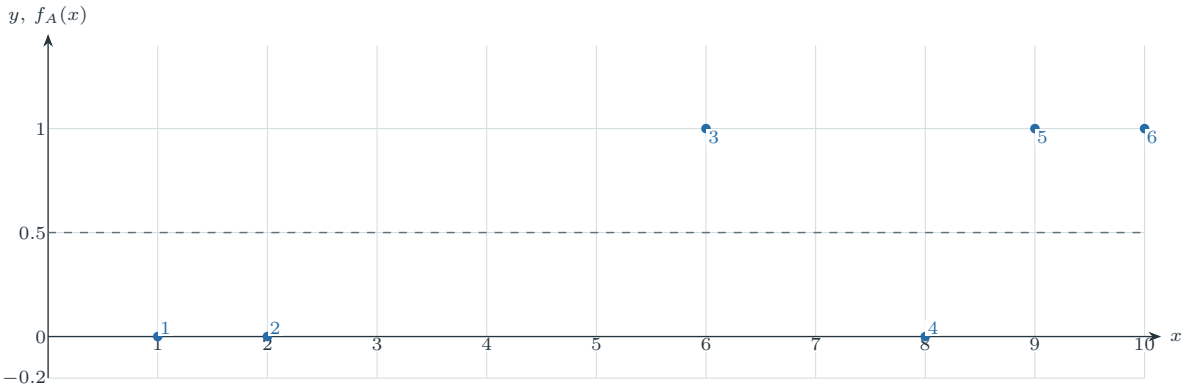
$y \in \{____, ____\}$ Positive Class $y =$ _____

2 Use line A, fitted to the first six examples, to calculate predictions and classes

$$f_A(x) = 0.1x - 0.1 \qquad \hat{y} = \begin{cases} 1 & f_A(x) \geq 0.5 \\ 0 & f_A(x) < 0.5 \end{cases}$$

Sensor i	x	y	$f_A(x)$	$\hat{y}$
1	1	0		
2	2	0		
3	6	1		
4	8	0		
5	9	1		
6	10	1		

3 Plot A's predictions over the actual data and connect them with a straight line



4 Find x where $f_A(x) = 0.5$ and count correct classifications among the first six examples

$0.1x - 0.1 = 0.5 \quad x =$ _____ Correct = _____/6

02 Limits of Linear Regression

Add the seventh sensor example $(x, y) = (27, 1)$ and refit the line using the same MSE

5 Compare predictions from A and the new line B at $x = 6$ and 27

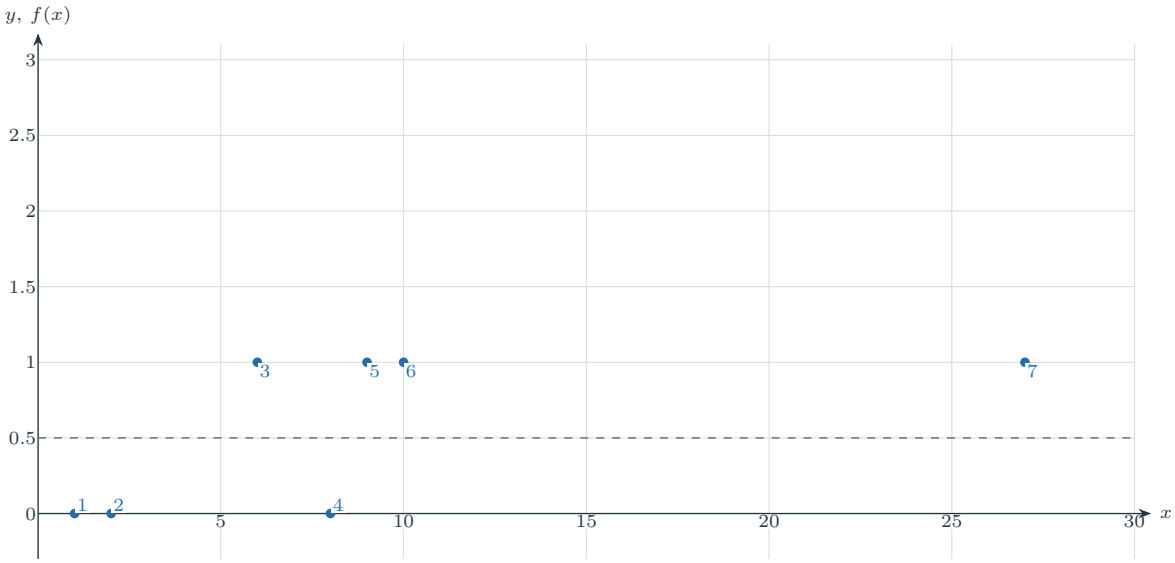
$f_A(x) = 0.1x - 0.1$ $f_B(x) = \dfrac{x+7}{28}$ $\hat{y} = 1$ if $f(x) \geq 0.5$

x	Target y	$f_A(x)$	$f_B(x)$
6	1		
27	1		

6 Calculate A and B at $x = 0, 10, 30$ and draw both lines

Model	$x = 0$	$x = 10$	$x = 30$
A			
B			

* Blue points: actual data A: solid line · B: dashed line



7 Explain how the class at $x = 6$ changed and whether all outputs can be interpreted as probabilities

$\hat{y}_A(6) =$ _____ $\hat{y}_B(6) =$ _____

03 Sigmoid Function

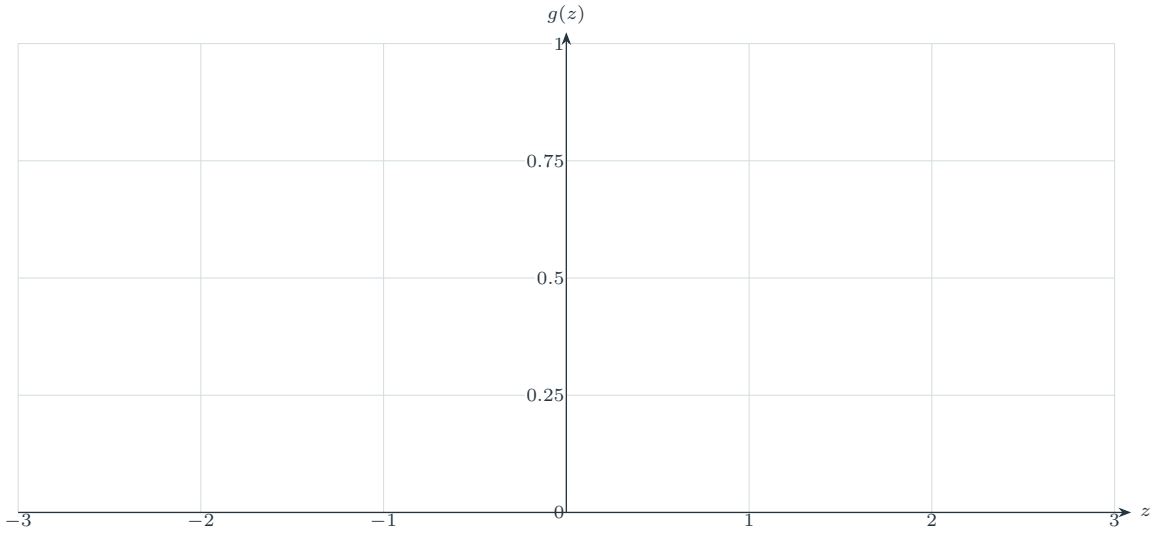
8 Use the supplied e^{-z} values to calculate $g(z)$

$$g(z) = \frac{1}{1+e^{-z}} \qquad e \approx 2.718 \qquad e^0 = 1$$

* Record three decimal places A calculator may be used for division and squares

z	e^{-z} (reference)	$g(z)$
-2	7.389	
-1	2.718	
0	1.000	
1	0.368	
2	0.135	

9 Plot $(z, g(z))$ and connect the points smoothly



10 Check how the input sign relates to the output and its range

$g(0) =$ _____ $z < 0 \Rightarrow g(z)$ _____ 0.5 $z > 0 \Rightarrow g(z)$ _____ 0.5

Calculate whether equal input changes of 1 produce equal output changes

$g(0) - g(-1) =$ _____ $g(2) - g(1) =$ _____

Can the output be exactly 0 or 1 for a finite z?

☐ Yes ☐ No

04 Logistic Regression

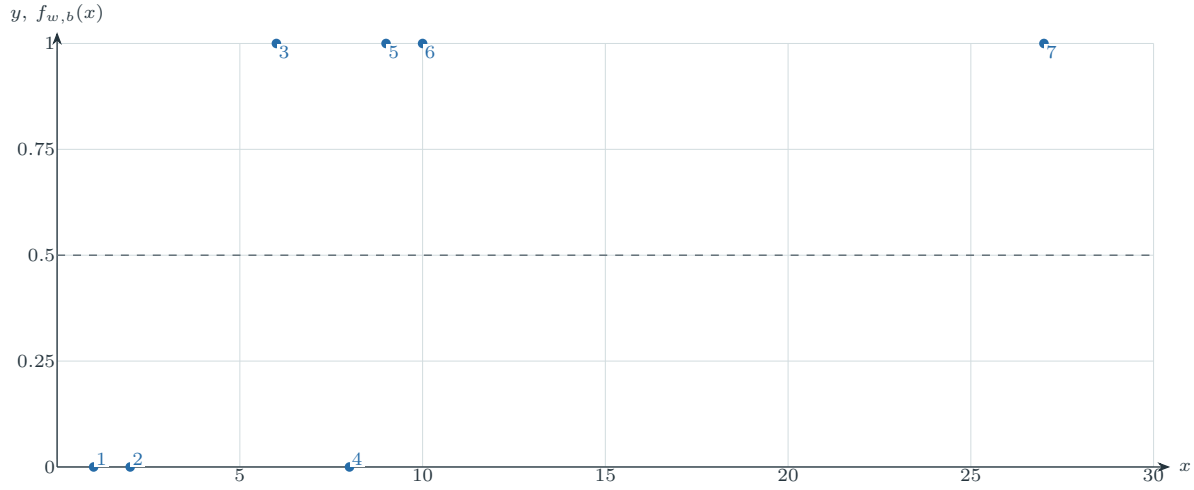
11 For all seven examples, calculate score z, probability p, then class $\hat{y}$

$$z = 0.25x - 1.5 \qquad p = f_{w,b}(x) = g(z) = \frac{1}{1+e^{-z}} \qquad \hat{y} = \begin{cases} 1 & p \geq 0.5 \\ 0 & p < 0.5 \end{cases}$$

$e^{1.25} \approx 3.4903$ $e^{1} \approx 2.7183$ $e^{0} = 1$ $e^{-0.5} \approx 0.6065$
$e^{-0.75} \approx 0.4724$ $e^{-1} \approx 0.3679$ $e^{-5.25} \approx 0.0052$

Sensor	x	y	z	p	$\hat{y}$
1	1	0			
2	2	0			
3	6	1			
4	8	0			
5	9	1			
6	10	1			
7	27	1			

12 Plot (x, p) over the actual data and draw the model curve



13 Interpret $p \approx 0.731$ at $x = 10$ and calculate the other class probability

$P(y = 1 \mid x = 10) \approx 0.731$ $P(y = 0 \mid x = 10) \approx$ _____

05 Decision Boundary

14 Find the input where the preceding model gives $p = 0.5$

$p = g(z) \geq 0.5 \iff z \geq 0 \qquad z = 0.25x - 1.5$

$0.25x - 1.5 = 0 \quad x =$ _____ $\hat{y} = 1$ if x _____ 6

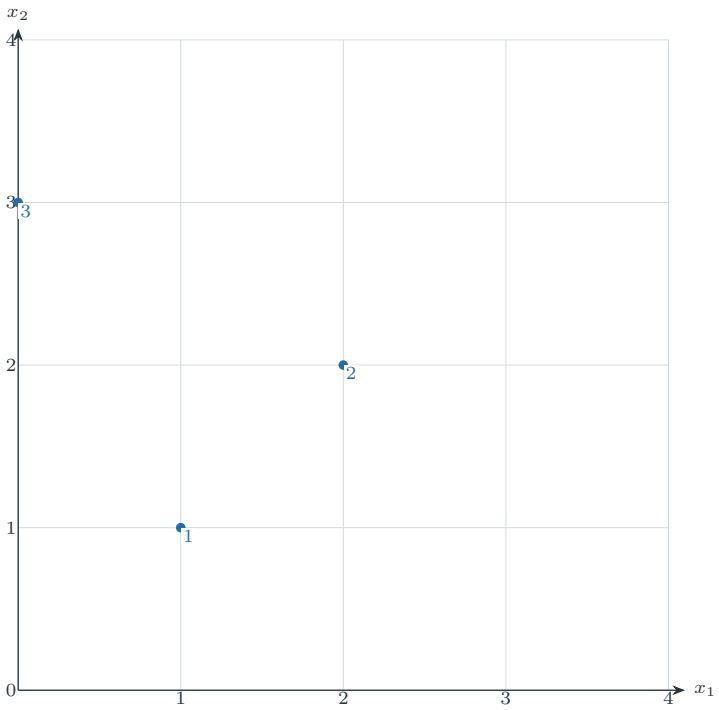
15 For two inputs, find both boundary intercepts and classify the following points

$z = x_1 + x_2 - 3 \qquad p = g(z)$ Decision Boundary: $z = 0$

$x_1 = 0 \Rightarrow x_2 =$ _____ $x_2 = 0 \Rightarrow x_1 =$ _____

(x_1, x_2)	z	p	$\hat{y}$
$(1, 1)$			
$(2, 2)$			
$(0, 3)$			

16 Connect the intercepts and label the regions with $\hat{y} = 0$ and 1



06 Linear Model + MSE

* Use all seven examples from here onward Fix $b = 0$ and vary only w to compare cost shapes

17 Substitute $f_w(x) = wx$ into the earlier MSE-based cost and check the expansion

$$J_{\text{linear}}(w) = \frac{1}{2m}\sum_{i=1}^{m}(wx^{(i)} - y^{(i)})^2 = \frac{1015w^2 - 104w + 4}{14} \qquad m = 7$$

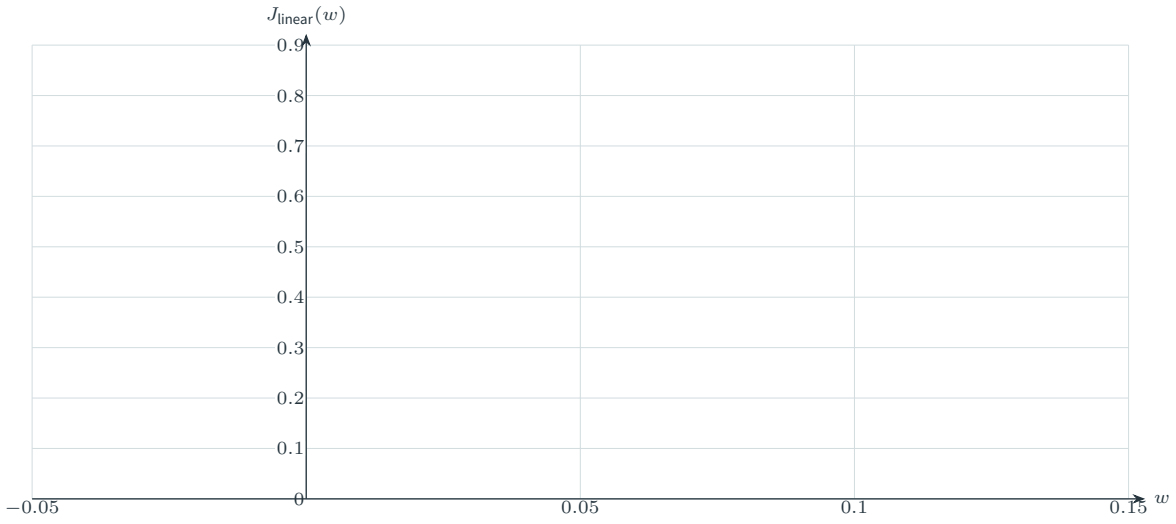
$$\sum_i (x^{(i)})^2 = 1015 \qquad \sum_i x^{(i)}y^{(i)} = 52 \qquad \sum_i (y^{(i)})^2 = 4$$

$(wx - y)^2 = w^2x^2 - 2wxy + y^2$ so $-2\sum_i x^{(i)}y^{(i)} =$ _____

18 Calculate the cost at the five values of w

w	Numerator $1015w^2 - 104w + 4$	$J_{\text{linear}}(w)$
-0.05		
0		
0.05		
0.1		
0.15		

19 Plot and connect $(w, J_{\text{linear}}(w))$, then select the lowest-cost w in the table



Selected $w =$ _____ Curve shape = _____

07 Sigmoid + MSE

20 Use the same MSE with $f_w(x) = g(wx)$ to calculate $J_{\text{MSE}}(0)$ and $J_{\text{MSE}}(0.2)$

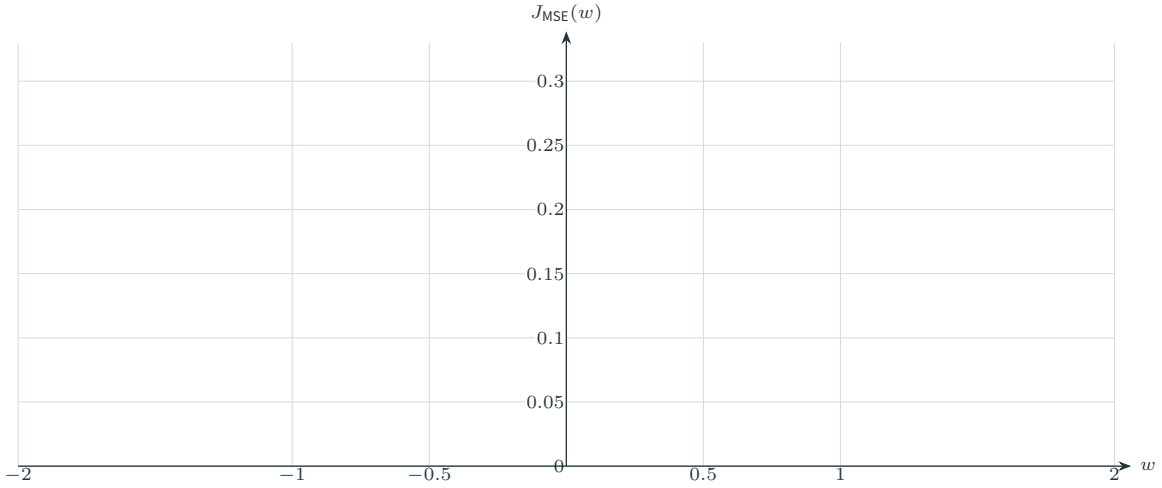
$$J_{\text{MSE}}(w) = \frac{1}{14}\sum_{i=1}^{7}\left(g(wx^{(i)}) - y^{(i)}\right)^2 \qquad b = 0$$

$w = 0 \Rightarrow g(0x) = 0.5 \quad J_{\text{MSE}}(0) = \dfrac{7 \times _____}{14} = _____$

x	1	2	6	8	9	10	27
y	0	0	1	0	1	1	1
$g(0.2x)$	0.5498	0.5987	0.7685	0.8320	0.8581	0.8808	0.9955
$(p-y)^2$							

21 Complete the cost table, plot all the points, and connect them

w	-2	-1	-0.5	0	0.2	0.5	1	2
$J_{\text{MSE}}(w)$	0.2868	0.2915	0.2920			0.1349	0.1650	0.1957



22 Join the points at $w = -2$ and 0 with a line segment and compare its midpoint with the cost at $w = -1$

If convex: $J\left(\dfrac{a+b}{2}\right) \leq \dfrac{J(a)+J(b)}{2}$

$\dfrac{0.2868 + 0.1250}{2} \approx$ _____ $J_{\text{MSE}}(-1) = 0.2915$ _____ 0.2059

08 Logistic Loss

23 Choose the formula for the actual target y and calculate the loss

$$p = f_{w,b}(x) \qquad L(p,y) = \begin{cases} -\log p & y = 1 \\ -\log(1-p) & y = 0 \end{cases}$$

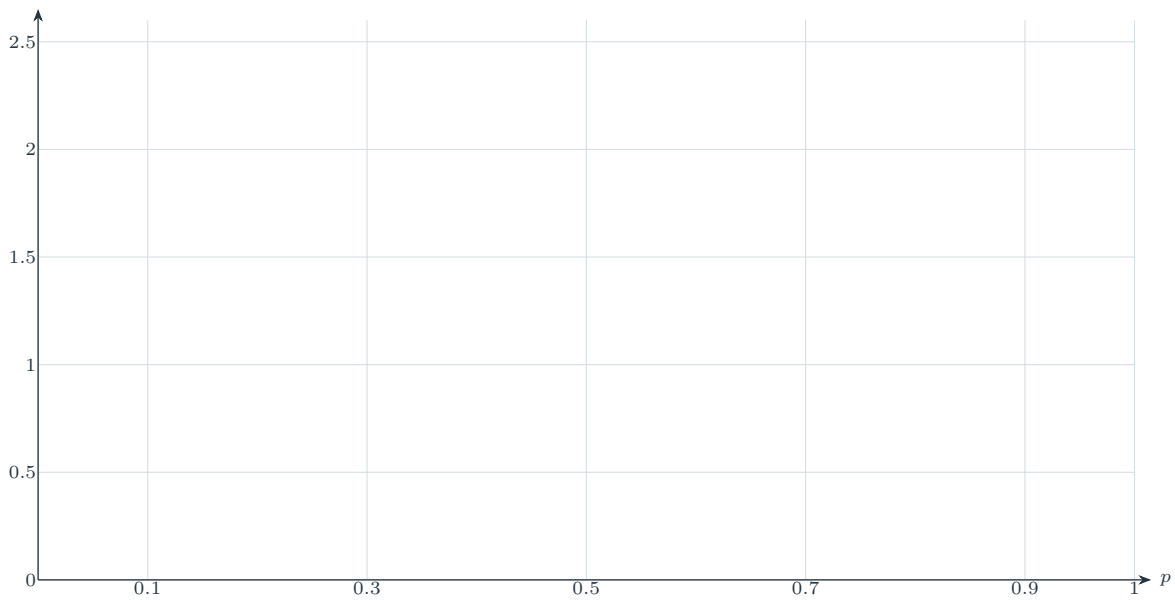
* Loss evaluates one example log denotes the natural logarithm

$\log(0.1) \approx -2.303$ $\log(0.5) \approx -0.693$ $\log(0.9) \approx -0.105$

p	$L(p,1)$	$L(p,0)$
0.1		
0.5		
0.9		

24 Plot both sets of loss values on one graph and draw their curves

* $y = 1$: solid line; $y = 0$: dashed line Do not plot infinite values at $p = 0$ or 1



25 For actual target 0, which prediction has greater loss, $p = 0.1$ or 0.9? Explain why

09 Logistic Cost / Convex

26 Average the losses over the same seven examples to find $J_{\log}(0)$ and $J_{\log}(0.2)$

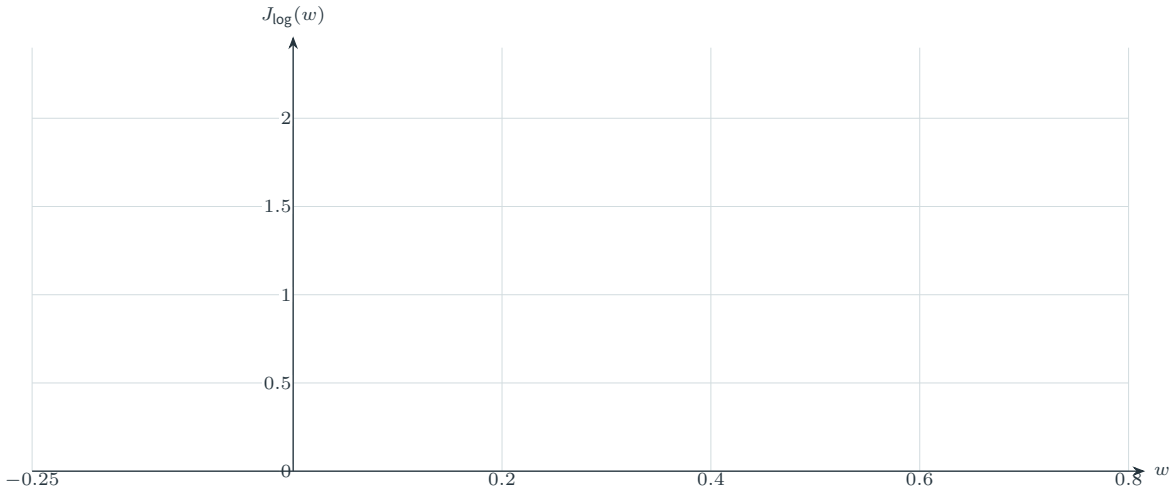
$$J_{\log}(w) = \frac{1}{7}\sum_{i=1}^{7} L(g(wx^{(i)}), y^{(i)}) \qquad b = 0 \qquad \log(0.5) \approx -0.6931$$

$$J_{\log}(0) = \frac{7 \times _____}{7} \approx _____$$

x $(w = 0.2)$	1	2	6	8	9	10	27
y	0	0	1	0	1	1	1
$\log p$	-0.5981	-0.5130	-0.2633	-0.1839	-0.1530	-0.1269	-0.0045
$\log(1-p)$	-0.7981	-0.9130	-1.4633	-1.7839	-1.9530	-2.1269	-5.4045
$L(p, y)$							

27 Plot and connect the values below; compare the curve with the preceding Sigmoid + MSE curve

w	-0.25	0	0.1	0.2	0.4	0.6	0.8
$J_{\log}(w)$	2.0798		0.5529		0.7794	1.0490	1.3380



28 Compare the middle cost with its neighbors at equally spaced $w = 0, 0.2, 0.4$

$$J_{\log}(0.2) \approx 0.5775 ____ \frac{0.6931 + 0.7794}{2} \approx _____$$